

PPS 22.1 - Radial Fields, Coulomb's Law

1. D: 5.85 (5.9) N

2. D: $\frac{(3.2 \cdot 10^{-19}) \cdot (1.6 \cdot 10^{-19})}{4 \cdot \pi \cdot 8.85 \cdot 10^{-12} \cdot 0.002^2}$

3. D

4. D

5.

6. a. B

b. $\frac{1}{49} \cdot 4.5 \cdot 10^{-19} = \frac{1}{9} \cdot x$

$$\frac{9}{49} \cdot 4.5 \cdot 10^{-19} = 8.265 \cdot 10^{-20}$$

7. a. Left, towards hydrogen nucleus.

b. $r = 2.5 \cdot 10^{-10}$ so

$$\begin{aligned}
 E_H &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1.6 \cdot 10^{-19}}{(2.5 \cdot 10^{-10})^2} \\
 &= 8.99 \cdot 10^9 \cdot 2.56 \\
 &= 2.301 \cdot 10^{10}
 \end{aligned}$$

$$\begin{aligned}
 E_I &= \frac{1}{4\pi\epsilon_0} \cdot \frac{-1.6 \cdot 10^{-19}}{(2.5 \cdot 10^{-10})^2} \\
 &= 8.99 \cdot 10^9 \cdot -2.56 \\
 &= -2.301 \cdot 10^{10}
 \end{aligned}$$

The magnitude of the resultant electric field strength is the difference between both fields.

$$= 2.301 \cdot 10^{10} - (-2.301 \cdot 10^{10})$$

$$= 5.6 \cdot 10^{10}$$

8. a. $r = 1.0 \cdot 10^{-15} \text{ m}$

$$\begin{aligned} F &= \frac{1}{4\pi\epsilon_0} \cdot \frac{Qq}{r^2} \\ &= 8.99 \cdot 10^9 \cdot \frac{(1.6 \cdot 10^{-19})^2}{(1.0 \cdot 10^{-15})^2} \\ &= 230 \text{ N} \end{aligned}$$

Same force from both protons, but they have different directions. The difference in direction $= 30^\circ$.

So horizontal components of each vector cancel. So the proton experiences a vertical force.

$$F = 2 \cdot (230 \cos(30^\circ)) = 398.4 \text{ N}$$

b.

C is equidistant from all protons. This means that at C, the magnitude of each individual electric field is the same from all three protons.

From the vector diagram, we see that if we draw the force vectors that C experiences tip to tail, they start and end at the same point.

This means that the force sums to zero. So the resultant electric field strength at C is zero.

9. From $-1.0 \cdot 10^{-12} \text{ C}$ charge:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \cdot \frac{-1.0 \cdot 10^{-12}}{0.1^2} \\ &= -0.899 \end{aligned}$$

From the $1.0 \cdot 10^{-12} \text{ C}$ charge:

$$\begin{aligned} E &= \frac{1}{4\pi\epsilon_0} \cdot \frac{1.0 \cdot 10^{-12}}{0.2^2} \\ &= 0.22475 \end{aligned}$$

$$-0.899 + 0.22475 = -0.674 \text{ N C}^{-1}$$

$$-0.674 \approx 0.7 \text{ N C}^{-1}$$

10.

$$0.674 = 8.99 \cdot 10^9 \cdot \frac{0.5 \cdot 10^{-12}}{r^2}$$
$$r = \sqrt{8.99 \cdot 10^9 \cdot \frac{0.5 \cdot 10^{-12}}{0.674}}$$
$$= 8.2 \text{ cm}$$

$$8.2 + 20 = 28.2 \text{ cm}$$